

Latent Bridge Matching: IDT-Calibrated Latent Transport for Domain-Level Artistic Transfer

Anonymous submission

Abstract

Domain-level artistic transfer has a blunt evaluation failure: when the source is already art, the unchanged source can obtain high target-style affinity. On the CLIP-separated WikiArt Distinct5 stress split, the reproduced SaMAM run remains below the identical-image transfer-only CLIP-S floor through its last trusted 2250-step checkpoint. We introduce *Latent Bridge Matching* (LBM), a 3.9M-parameter latent transport family built from training-side endpoint pairing, a style-conditioned residual field, and terminal distribution matching in VAE latent space, keeping style-specific customization in a small-model learning regime rather than adapting a large generative prior. On Distinct5-512, LBM forms a controllable operating-

point frontier rather than a single conservative checkpoint: compact anchors clear the IDT floor within 1.2 minutes on an RTX 3060, structure-aware and Pattern+Stokes successors push transfer CLIP-S into the 0.72–0.73 band, and the family spans distinct low-damage, Pareto-knee, and style-ceiling regimes under the same packet. On the historical strict-750 reference surface, LBM reaches 0.7161 CLIP-S / 0.4514 LPIPS at 114 ms/image. The practical implication is direct: IDT calibration prevents false stylization claims, and compact latent transport keeps customized style modeling in a lightweight consumer-GPU regime instead of a large-prior adaptation workflow.

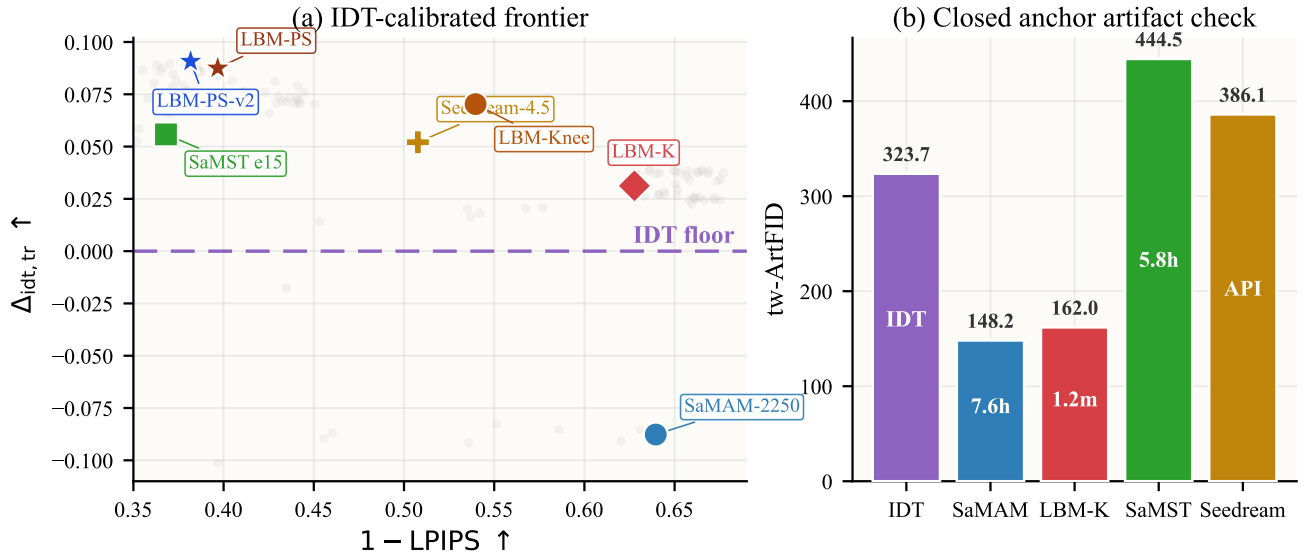


Figure 1: IDT-calibrated Distinct5-512 summary under the shared packet. Panel (a) plots selected operating points in transfer-only $\Delta_{idt, tr}$ versus $1 - \text{LPIPS}$ space, with faint gray dots showing the broader recorded development landscape. Conservative compact anchors remain near the low-damage regime, later successors move upward into a stronger style band, and Seedream-4.5 is shown as an external large-prior reference rather than a style-ID model. Panel (b) reports the closed *transfer-only* targetwise ArtFID comparison for compact anchors, reproduced baselines, and the Seedream reference.

Introduction

Domain-level artistic transfer asks a model to render a source image according to a target style identity rather than a target style exemplar. This setting exposes a simple but serious evaluation failure: when the source image is already artwork, the unchanged source can obtain high target-style affinity. A

method may therefore appear competitive by preserving an art-domain prior, or by making visible but misdirected edits, without actually moving toward the requested target domain. We use the identical-image transfer (IDT) floor to make this failure explicit and report signed target-style movement above that floor.

Our target regime is also narrower and more practical than reference-guided or diffusion-prior stylization. At inference time, the model receives only a source image and a target style identifier; it should not require a style image, per-image optimization, or adaptation of a large generative prior. This makes the problem a low-cost customization problem as much as a stylization problem: the desired model should expose selectable style-content operating points while remaining compact enough to retrain on consumer hardware.

We propose *Latent Bridge Matching* (LBM), a compact latent transport family in VAE space. In the current Distinct5 mainline, LBM combines training-side endpoint pairing, a style-conditioned latent field, and terminal SA-SWD with kinetic motion control. The resulting family forms a controllable IDT-calibrated frontier rather than a single conservative checkpoint: compact anchors clear the IDT floor in a low-damage regime, while StructOT and Pattern+Stokes successors move into a stronger high-style band under the same style-ID inference contract. This framing is what distinguishes the paper from both reference-guided stylization and large-prior adaptation lines (Gatys, Ecker, and Bethge 2016; Huang and Belongie 2017; Liu et al. 2024, 2025; Zhang et al. 2024; Chung, Hyun, and Heo 2024; Zhou et al. 2024; Yeh et al. 2020; Wright and Ommer 2022).

Our contributions are threefold.

- **Evaluation diagnostic.** We introduce an IDT-calibrated protocol for domain-level art-to-art transfer, reporting transfer-only Δ_{IDT} so that target-style movement is separated from the unchanged art-domain prior.
- **Method family.** We propose LBM, a compact style-ID latent transport family that separates training-side endpoint pairing, style-conditioned latent execution, kinetic motion control, and terminal SA-SWD matching in VAE space.
- **Frontier evidence.** We show that LBM is not a single conservative checkpoint but a controllable operating-point frontier on Distinct5-512: compact anchors clear the IDT floor at minute-scale training cost, while StructOT and Pattern+Stokes successors enter a substantially stronger style regime under the same packet.

Related Work

Arbitrary style transfer began with optimization-based feature reconstruction and Gram statistics (Gatys, Ecker, and Bethge 2016), then moved toward feed-forward alignment and local correspondence through AdaIN, SANet, ArtFlow, AdaAttN, StyTr2, and related variants (Huang and Belongie 2017; Park and Lee 2019; An et al. 2021; Liu et al. 2021; Deng et al. 2022). More recent systems refine style injection through aesthetic guidance, semantic-region control, or stronger pretrained priors (Hong et al. 2023; Huang et al. 2024; Zheng et al. 2024; Zhang et al. 2025; Shang et al. 2025; Zhang et al. 2024; Chung, Hyun, and Heo 2024; Deng et al. 2024). These methods mainly ask how a style exemplar or large prior should be used at inference. Our setting is different: inference receives only a target style identity, so the main question is whether a compact style-side representation remains executable after passing through a content-conditioned

latent renderer.

The closest comparison line is efficient multi-style and compact style-ID transfer. SaMST studies style modeling and transfer through pluggable style representations (Liu et al. 2024), while Mamba-ST and SaMam revisit global mixing through state-space backbones (Botti et al. 2025; Liu et al. 2025). LBM does not claim a new global-mixing primitive. Its contribution is training-side and objective-side: pairing-cache endpoint selection, compact latent transport, and terminal distribution matching under a style-ID interface. On the evaluation side, prior work already notes that automatic style-transfer metrics are inconsistent and can misalign with human judgment (Zhou et al. 2024; Yeh et al. 2020; Wright and Ommer 2022). This motivates our Distinct5 use of the unchanged-image floor and Δ_{idt} as explicit diagnostics for whether a method actually moves toward the requested target style.

Method

Overview

LBM is a latent transport family in VAE space for domain-level artistic transfer. For the Distinct5-512 packet, the source latent is $z_0 \in \mathbb{R}^{4 \times 64 \times 64}$; the historical strict-750 packet uses $4 \times 32 \times 32$ latents and is retained only as secondary support. Let s denote a target style identifier. The active Distinct5 line combines four pieces: (i) training-side endpoint pairing that selects candidate target latents without paired supervision; (ii) a style-conditioned latent field that executes the requested motion; (iii) kinetic regularization that discourages unnecessary displacement; and (iv) terminal distribution matching that checks whether the resulting endpoint has target-style patch statistics. At inference, the model receives only the source image and target style identifier; target-domain latents remain training-side supervision only.

In the reproduced Distinct5 mainline, the active headline rows use endpoint-mode OMF training rather than the earlier random-time flow residual. Writing

$$\hat{v} = v_\theta(z_0, 1, s), \quad \hat{z}_1 = z_0 + \hat{v}, \quad (1)$$

the active objective is

$$\mathcal{L}_{\text{active}} = \lambda_{\text{term}} \mathcal{L}_{\text{term}}(\hat{z}_1) + \lambda_{\text{kin}} \|\hat{v}\|_2^2, \quad (2)$$

where candidate terminal targets are selected by the pairing cache and the parent flow residual is disabled in the headline rows ($w_{\text{flow}} = 0$). We therefore describe the paper-facing Distinct5 family as pairing-cache / terminal-SWD / kinetic OMF rather than as an online Sinkhorn or random-time flow-matching result family.

Endpoint pairing, execution, and terminal matching

We train directly on precomputed VAE latents without paired supervision. For each source latent z_0 and target style s , the pairing cache exposes a selector-restricted candidate set $\mathcal{C}(z_0, s)$ from the target domain. The terminal target $\hat{z}_1$ used by the loss is drawn from that scheduled set. This pairing step determines where supervision should pull, but not how much motion is actually executed.

Latent Bridge Matching (LBM)

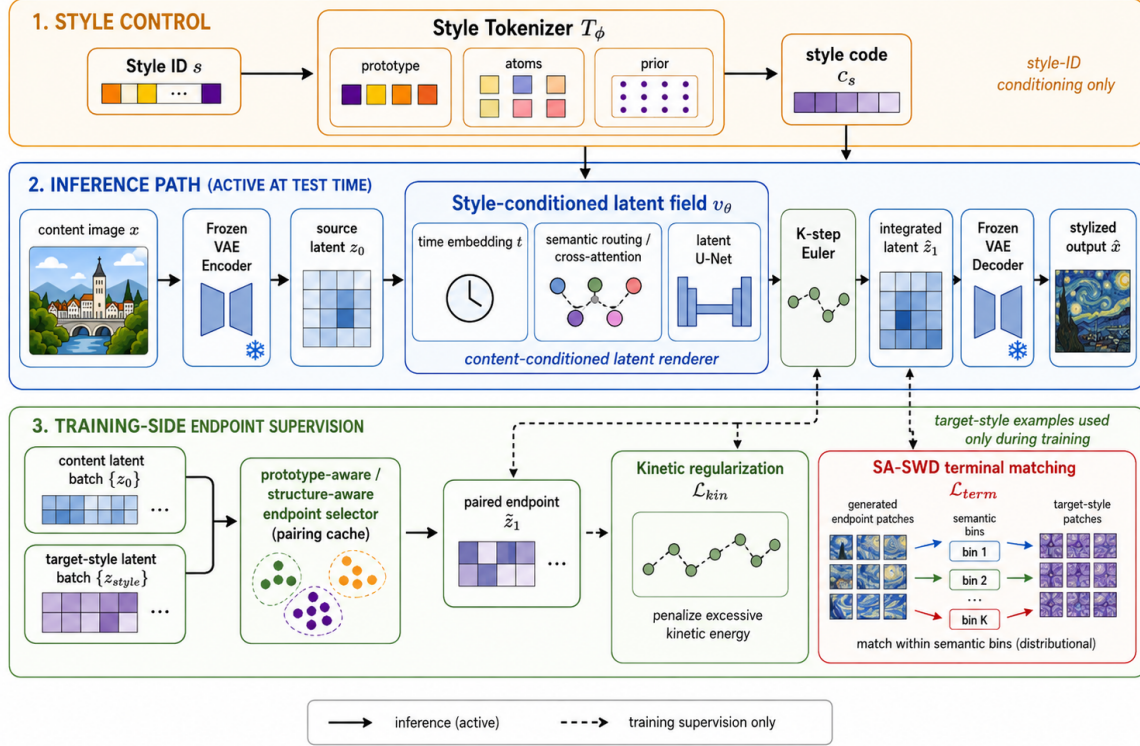


Figure 2: LBM combines a style tokenizer, a content-conditioned latent renderer, and training-side endpoint supervision with terminal SA-SWD.

Execution is handled by a time-conditioned latent backbone with sinusoidal time embeddings, style spatial priors, and semantic cross-attention. The style-side interface is separated from the renderer: the tokenizer receives only a target style identifier and produces a compact control code c_s , while the renderer decides how much latent motion is needed under the current content. In the current mainline this interface is paired with prototype-aware target queues and content-guided routing, which empirically improve executed style-content tradeoffs more reliably than simply enlarging the style code.

The endpoint is then regularized by semantic-aligned sliced Wasserstein distance. Let $\Phi_\theta(z_0, s)$ denote the Euler-integrated endpoint. The terminal objective is

$$\mathcal{L}_{\text{term}} = \text{SWD}_1(\Phi_\theta(z_0, s), \tilde{z}_1), \quad (3)$$

where sorted one-dimensional projections compare generated endpoint patches against target-style patches without requiring spatial correspondence. In the current SA-SWD instantiation, projection directions are chosen from the same routing signal that injects style into the backbone. This design keeps target-style pressure distributional rather than point-wise, which is important in compressed VAE latents where direct endpoint MSE encourages mean solutions. Bounded design-grounding analysis for kinetic regularization, Euler discretization, and the historical OT-style parent assignment

map is retained as supplementary support rather than consuming main-paper space.

Operating-point variants. All reported LBM rows share the same style-ID inference contract: inference receives only the source image and target style identifier. LBM-K is the compact earlier-mainline anchor. LBM-Knee is the promoted low-LPIPS successor that combines the stronger successor family with additional queue- and anisotropic/Stokes-style regularization to preserve more content while staying above the IDT floor. LBM-PS is the balanced high-style successor, and LBM-PS-v2 is the style-ceiling variant with weaker viscosity. These rows therefore do not define different inference problems; they expose different operating points under the same compact latent transport interface.

Experiments

Protocol hierarchy and metrics

We organize the evidence into two non-interchangeable result families, with Distinct5-512 as the primary stress benchmark and the historical strict-750 packet as contextual support. This separation is necessary because the same scalar pair, CLIP-style and LPIPS, can be misleading when datasets, image resolution, or baseline training protocols change.

Historical strict-750. The main legacy protocol uses five domains—photo, Hayao, Monet, Van Gogh, and Cezanne—

and evaluates all 5×5 source-target directions with 30 source images per source domain, for 750 generated outputs. This is the protocol used for comparisons to SaMST, S2WAT, StyleID, AdaIN, StyTr2, CAST, AesFA, and AesPA-Net.

Distinct5-512 stress split. The new stress split uses Early Renaissance, Impressionism, Minimalism, Rococo, and Ukiyo-e, with 1000 training images and 30 held-out test images per class. These five classes were chosen as the most strongly separated subset under a CLIP-style screening pass over WikiArt categories, rather than being hand-picked for LBM. Evaluation again uses all 5×5 directions over 750 generated images. This split complements the historical table with a stronger domain-separation test and an explicit identical-image baseline.

We report CLIP-style (CLIP-S) and CLIP-content (CLIP-C) using CLIP ViT-B/32 embeddings (Radford et al. 2021), together with LPIPS-content and

$$EC = \text{CLIP-style} \cdot (1 - \text{LPIPS}). \quad (4)$$

For Distinct5-512 we additionally report style gain over the unchanged-image reference,

$$\Delta_{\text{idt}} = \text{CLIP-S}(\text{method}) - \text{CLIP-S}(\text{idt}), \quad (5)$$

under both the full 5×5 protocol and the transfer-only off-diagonal subset. We use this quantity to define the metric-stress regime exposed by Distinct5-512: raw CLIP-style should not be treated as sufficient evidence of successful target-style transfer if a method either fails to exceed the identical-image prior ($\Delta_{\text{idt}} \leq 0$) or exceeds it only by moving into a clearly damaged LPIPS region while also deteriorating broad artifact diagnostics. In this paper, CLIP-S is interpreted as target-style affinity, LPIPS as content displacement, and targetwise ArtFID as a target-domain plausibility and artifact diagnostic rather than a direct style-gain score. Aggregate ArtFID is kept only as an auxiliary art-domain/identity diagnostic outside this main Distinct5 comparison. In particular, lower targetwise ArtFID alone does not certify useful stylization on separated art-to-art transfer: its content term rewards source preservation, and its distributional term can reward remaining in a plausible target-domain art manifold even when the requested target-style movement is still below the unchanged-image baseline. EC is used only as a Pareto summary; claims are checked against CLIP-S and LPIPS individually. For artifact-sensitive analysis we also report MUSIQ, MANIQA, DISTS-content, high-frequency patch KID, FFT slope error, shallow Gram micro statistics, and KID where available. All strict-750 baselines use the same source images, target domains, preprocessing, and identity directions. For arbitrary-style methods, one target-domain reference image is supplied per target style; for style-id LBM, the model receives only the target domain identifier. The ‘idt’ control is introduced here as a split-specific diagnostic baseline for art-to-art transfer: our claim is that Distinct5 makes explicit the role of the unchanged-image prior on this split, not that any single existing metric should be discarded or universally replaced.

Distinct5-512 stress benchmark

Distinct5-512 is an IDT-calibrated operating-point benchmark, not a single-score leaderboard. All five domains come

from standard WikiArt classes and are selected because they are strongly separated under a CLIP-style screening pass. In this regime, unchanged output is no longer a plausible success proxy: the *idt* baseline already reaches CLIP-S 0.6801 while doing no transfer at all. Signed target-style movement is therefore the first sanity check. Under the shared full 5×5 / 750-output protocol, the earlier compact anchors LBM-F/K establish a conservative low-damage regime: LBM-F reaches transfer CLIP-S 0.6644 with LPIPS 0.3245 after 1.2 minutes of checkpoint training, and LBM-K raises style to 0.6712 at LPIPS 0.3723 while remaining well below the SaMST damage band. The more important result is that the family does not stop there. A promoted content-preserving successor now reaches 0.7102 transfer CLIP-S at 0.5397 content preservation ($1 - \text{LPIPS}$), while the higher-style successors push the frontier to 0.7274 / 0.3967 and 0.7307 / 0.3817 in transfer CLIP-S / ($1 - \text{LPIPS}$). This is not a marginal +0.02 style story. It is a family-level frontier with three practically different regimes: conservative low-damage anchors, a content-preserving successor knee, and a style-ceiling operating point.

To keep this frontier paper-safe rather than post hoc, we fix representative rows before qualitative inspection. The manuscript table reports one compact anchor from the earlier mainline, one lower-LPIPS successor tradeoff point, one balanced high-style point, and one style-ceiling point. The broader development landscape is shown only as faint background support in Figure 1, not as a second leaderboard.

The reproduced baselines remain important because they bound those regimes. SaMST-512 e15 reaches 0.6957 / 0.6319 and can produce locally strong target-style jumps, but it does so in a clearly higher-damage region and with targetwise ArtFID 395.7 on the full scope and 444.5 on the transfer-only scope; the retained e5/e15 packet is already near a plateau on transfer CLIP-S and LPIPS, with less than 0.004 / 0.002 difference between the checkpoints. Seedream-4.5 provides a stronger external large-prior reference under a fixed API protocol: we use the source image together with a fixed target-domain prompt template of the form “convert the image to *<style>* style,” without retraining or style-ID control. Its transfer row is 0.6920 / 0.4923, which is better than the compact anchor but still below the promoted LBM knee point on both style gain and content preservation. At the same time, Seedream keeps the strongest identity row among the currently compared points, which is why we treat it as a useful outer reference rather than as a same-interface baseline. At the trusted SaMAM 2250 checkpoint, transfer CLIP-S remains below the IDT floor even though targetwise ArtFID improves to 146.1 on the full scope and 148.2 on the transfer-only scope. This is the evaluation failure exposed by Distinct5: lower targetwise ArtFID can still coincide with zero or negative target-style gain when a method preserves source structure and stays inside a plausible target-art manifold without moving toward the requested target style. Distinct5 therefore turns the paper’s evaluation claim into an operating-point test within this split: target-style movement should be demonstrated relative to IDT, and a paper-facing result should be reported as a selected Pareto point rather than as one arbitrarily chosen checkpoint.

Table 1: Selected Distinct5-512 operating points under the shared transfer-only packet. ‘idt’ is the unchanged-image floor. Rows are chosen to represent distinct roles on the current frontier rather than a single checkpoint family: a compact conservative anchor, a content-preserving successor knee, two higher-style promoted successors, and one external large-prior reference. We report only transfer-scope metrics here so the main table does not mix all-pairs and transfer-only readings.

Point	Role	CLIP-S _{tr} ↑	1 - LPIPS ↑	$\Delta_{\text{idt}, \text{tr}}$ ↑	Closure
idt	no-op floor	0.6399	1.0000	0.0000	closed
SaMAM-2250	compact baseline	0.5523	0.6395	-0.0877	closed
SaMST e15	high-style baseline	0.6957	0.3681	+0.0558	closed
Seedream-4.5	large-prior ref.	0.6920	0.5077	+0.0521	closed
LBM-K e1	conservative anchor	0.6712	0.6277	+0.0312	closed
LBM-Knee e13	content-preserving successor	0.7102	0.5397	+0.0703	frontier
LBM-PS e13	balanced high-style	0.7274	0.3967	+0.0875	frontier
LBM-PS-v2 e13	style ceiling	0.7307	0.3817	+0.0908	frontier

Table 2 isolates the recorded operating-point cost under the same Distinct5 packet, but only for rows whose timing provenance is already fully closed. The broader successor family should therefore be read as a quality frontier first, not as a normalized timing leaderboard: the later Pattern+Stokes and structure-aware OT packets strengthen the style side of the frontier, while compact-anchor timing remains the cleanest accessibility evidence. The qualitative interpretation is consistent with the table-level result: SaMST can produce locally strong target-style jumps, but those gains sit in a much higher-damage region, Seedream-4.5 remains an out-of-interface large-prior reference rather than a style-ID baseline, and SaMAM still often edits without producing reliable target-style movement above the no-op floor.

The same reading survives an explicit three-scope check on the promoted content-preserving successor. A local retest of the selected e13 knee point gives transfer 0.7101 / 0.4604, all-pairs 0.7302 / 0.4560, and identity 0.8105 / 0.4384 in CLIP-S / LPIPS. For context, Seedream-4.5 gives transfer 0.6920 / 0.4923, all-pairs 0.7198 / 0.4767, and identity 0.8312 / 0.4145. The promoted LBM successor therefore improves target-style movement and content preservation over this large-prior reference on the transfer and all-pairs scopes, while Seedream keeps the strongest identity row. We do not use this comparison to claim same-interface dominance, since Seedream is not a compact style-ID latent renderer; we use it only as additional evidence that the promoted successor frontier is not a trivial CLIP-only artifact.

A paired bootstrap over the 600 IDT-aligned transfer rows gives 95% intervals of [0.0210, 0.0280] for LBM-F and [0.0273, 0.0352] for LBM-K, confirming that their positive transfer-only Δ_{idt} values are stable within this reviewed split. The same bootstrap packet also keeps SaMST strictly above IDT, but only in the same high-LPIPS region summarized in Table 1. We use this bootstrap only as within-split stability evidence; it is not presented as a cross-split or human-preference validation.

Adaptation cost and accessibility. Figure 1 makes the practical gap visible without resorting to a synthetic normalized speedup claim. Conservative LBM anchors cross the

Table 2: Distinct5-512 recorded operating-point cost under the reviewed full 5×5 / 750-output packet. Reported train time is cumulative training wall-clock to the shown checkpoint, with evaluation excluded. This is recorded operating-point cost, not a normalized time-to-quality comparison.

Method	Recorded point	Train to point
LBM-F	e1	1.2 min
LBM-K	e1	1.2 min
SaMAM	step 2250	7.6 h
SaMST	e15	5.8 h

Note. Distinct5 times are cumulative training wall only, with evaluation excluded. We intentionally omit a shared inference column because same-grade pure-generation timing is still incomplete for the active manuscript rows SaMAM-2250 and SaMST-e15; leaving blanks would overstate closure. Post-2250 SaMAM timings are intentionally excluded.

Distinct5 IDT floor after 1.2 minutes of training on a single RTX 3060, whereas the trusted SaMAM-2250 checkpoint remains below that floor after 7.6 hours and SaMST exceeds it only after 5.8 hours in a much higher-damage region. The safe reading is therefore an evidence-closed compact-anchor claim rather than a universal time-to-parity theorem: within this domain-style setting, compact latent transport can shift customized stylization from a resource-intensive adaptation problem toward a minute-scale small-model learning problem.

Historical support and secondary evidence

We keep the historical strict-750 packet only as contextual support rather than as a second main benchmark. On that older surface, LBM remains a competitive compact retrainable point with 0.7161 CLIP-S / 0.4514 LPIPS at 114 ms/image, while the strongest reproduced competing rows either collapse content or enter noisier artifact regimes. The key historical value for the current paper is therefore qualitative and bounded: it shows that the compact transport story is not unique to Distinct5 and that the method can still occupy a cleaner operating region on an older mixed-family surface.

The same policy applies to ablations. Tokenizer and representation studies show that raw code capacity alone is not the bottleneck: content-guided execution and prototype-aware target queues improve the frontier more reliably than larger undifferentiated style codes. Mechanism ablations show that terminal distribution matching is the primary style driver, while kinetic regularization acts as a practical path stabilizer in the current OMF / field regime. We keep the detailed tables, artifact-sensitive diagnostics, and extended sensitivity sweeps as secondary evidence rather than spending main-paper pages on them; their role is to bound the interpretation of the Distinct5 frontier, not to replace it.

All reported evaluations use fixed seeds, stored resolved configurations, source snapshots, checkpoints, and per-epoch logs. Historical timings were measured on an RTX 4070 Laptop GPU unless otherwise noted. For Distinct5-512, Table 2 records the evidence-closed operating-point times. Across

both packets, the intended takeaway is practical rather than leaderboard-style: meaningful style-id customization can remain in a minute-to-hour consumer-GPU regime without adapting a large pretrained generator.

Discussion and Limitations

The main lesson is that style transfer quality cannot be inferred from raw CLIP-style alone. On the historical strict-750 protocol, StyleID has the highest raw style but collapses content, SaMST is slightly stronger than LBM on CLIP-S and CLIP-C but shows grain-like local artifacts, and LBM occupies a cleaner LPIPS/artifact-sensitive operating point. Distinct5-512 sharpens the same lesson further because its no-op baseline is measured explicitly: higher raw style is achievable, but only by moving into a severe LPIPS and targetwise ArtFID failure region, and some baselines fail even to exceed the identical-image prior. This exposes a community-relevant blind spot in art-to-art evaluation: preserved art-domain priors can masquerade as target-style success unless the unchanged-image floor is reported. Earlier AST evaluation work already notes that protocols are inconsistent across papers and that automatic scores can be style-dependent or misaligned with human judgment (Zhou et al. 2024; Yeh et al. 2020; Wright and Ommer 2022). Our claim is therefore concrete but local: on Distinct5-512, the unchanged-image reference and Δ_{idt} act as useful diagnostics because they agree with the observed failure of sub-IDT baselines to move consistently toward the requested target style.

These results change the role of LBM from a single operating point to a selectable style-content family. The current successor packet shows that compact latent transport spans a measurable Pareto frontier with conservative anchors in the low-damage region, Pattern+Stokes successors in the 0.72–0.73 transfer CLIP-S band, and lower-LPIPS promoted side lines clustered around the mid-0.55 LPIPS region while remaining above 0.71 style. This changes the paper’s performance story from a modest +0.02 / +0.03 IDT-adjusted gain to a family that can directly challenge the reproduced high-style baseline while keeping multiple operating regimes available. The practical implication is separate from that metric point: the same experiments show that meaningful style-id customization can be trained and deployed in a lightweight small-model regime, lowering the barrier for researchers and creators who want customized stylization without maintaining a large generative backbone. We therefore treat adaptation cost as community-relevant evidence rather than an appendix-side detail and encourage future domain-style papers to report selected operating points, target-style movement, and artifact-sensitive checks together rather than relying on one raw style score.

The tokenizer experiments narrow the design question. Within the tested Distinct5 design space, raw code capacity alone is not the bottleneck. The sharper issue is whether target-style information survives execution through a content-conditioned vector field strongly enough to yield real style gain beyond the unchanged-image reference. The current evidence therefore favors target-style carriers, prototype-aware target queues, and content-aware routing over undifferentiated token expansion. Matched successor probes rein-

force the same point: tokenizer code separation maps only moderately into executed separation, while style gain over the unchanged-image reference tracks executed movement more closely than raw code geometry. This keeps executed style survival as the sharper mechanism question in the current successor family, and it suggests that the next representation stage should factor target representation from execution budget rather than simply adding parameters.

The study has several limitations. First, the main protocol is domain-level rather than reference-image-specific, so it does not attempt to reproduce a unique style image. Second, Euler integration is deliberately simple; although the step sweep is stable in the reported regime, more aggressive paths or higher resolutions may need better solvers or stronger constraints. Third, SA-SWD is an adaptive-projection distribution loss; current controls show that most of its benefit comes from the distributional term rather than from the specific semantic projection rule. Fourth, artifact-sensitive metrics are still proxies for visual preference; human studies or rubric-based VLM assessment would strengthen the claim further. In particular, the promoted successor rows currently have transfer-style / LPIPS closure, while full successor artifact-sensitive and human-preference closure remains future work. Finally, Distinct5-512 is a five-domain stress split rather than a universal art benchmark, so broader domain coverage remains future work even though the current cross-method comparison is already informative on this split; we therefore do not claim universal adoption of IDT-style diagnostics without broader validation.

Conclusion

We presented Latent Bridge Matching, a compact latent transport design for domain-level artistic transfer. The method separates training-side endpoint selection, path-wise field learning, adaptive-projection terminal distribution matching, and style-tokenized execution. Under the reproduced historical strict-750 protocol, LBM delivers a favorable reproduced balance of style strength, content fidelity, artifact control, and practical footprint within the mixed-family comparison packet considered here. The Distinct5-512 stress benchmark sharpens the paper’s main evaluation point: once the identical-image prior is made explicit, target-style movement must be read against IDT rather than from raw CLIP-S or ArtFID alone. On that split, LBM is not merely a safe low-damage alternative. It is a controllable family of operating points: conservative compact anchors clear the IDT floor early, structure-aware and later Pattern+Stokes successors move into a stronger high-style regime, and the full family spans a measurable style-content frontier instead of a one-point improvement claim.

Just as importantly, the reproduced minute-scale RTX 3060 adaptation path and the 3.91M / 114 ms historical operating point show that customized stylization can remain a small-model learning problem rather than a large-prior adaptation problem, lowering the barrier for researchers and creators who want low-cost style-specific models on consumer GPUs. The tokenizer study then isolates the remaining mechanism question. Within the tested Distinct5 tokenizer variants, matched localization controls show that executor-side

refresh recovers more than style-side refresh alone on the current local surface, while the same-family H path-stability packet supports the bounded claim that kinetic regularization acts as a practical stabilizer in the current OMF / field regime. The next representation target is therefore to preserve target-style separation through execution itself, via explicit target carriers and tighter content-risk control.

References

- An, J.; Huang, S.; Song, Y.; Dou, D.; Liu, W.; and Luo, J. 2021. ArtFlow: Unbiased Image Style Transfer via Reversible Neural Flows. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.
- Botti, F.; Ergasti, A.; Rossi, L.; Fontanini, T.; Ferrari, C.; Bertozzi, M.; and Prati, A. 2025. Mamba-ST: State Space Model for Efficient Style Transfer. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 7786–7795.
- Chung, J.; Hyun, S.; and Heo, J.-P. 2024. Style Injection in Diffusion: A Training-free Approach for Adapting Large-scale Diffusion Models for Style Transfer. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.
- Deng, Y.; He, X.; Tang, F.; and Dong, W. 2024. Z*: Zero-shot Style Transfer via Attention Reweighting. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 6934–6944.
- Deng, Y.; Tang, F.; Dong, W.; Sun, W.; Huang, F.; and Xu, C. 2022. StyTr2: Image Style Transfer with Transformers. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.
- Gatys, L. A.; Ecker, A. S.; and Bethge, M. 2016. Image Style Transfer Using Convolutional Neural Networks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*.
- Hong, K.; Jeon, S.; Yang, H.; Fu, J.; and Byun, H. 2023. AesPA-Net: Aesthetic Pattern-Aware Style Transfer Networks. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*.
- Huang, X.; and Belongie, S. 2017. Arbitrary Style Transfer in Real-time with Adaptive Instance Normalization. In *Proceedings of the IEEE International Conference on Computer Vision*.
- Huang, X.; Zhang, Z.; Zhang, J.; et al. 2024. AesFA: An Aesthetic Feature-Aware Arbitrary Neural Style Transfer. In *Proceedings of the AAAI Conference on Artificial Intelligence*.
- Liu, H.; Wang, L.; Zhang, Y.; Yu, Z.; and Guo, Y. 2025. SaMam: Style-aware State Space Model for Arbitrary Image Style Transfer. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 28468–28478.
- Liu, S.; Lin, T.; He, D.; Li, F.; Wang, M.; Li, X.; Sun, Z.; Li, Q.; and Ding, E. 2021. AdaAttN: Revisit Attention Mechanism in Arbitrary Neural Style Transfer. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*.
- Liu, Y.; et al. 2024. Pluggable Style Representation Learning for Multi-Style Transfer. In *Proceedings of the Asian Conference on Computer Vision*.
- Park, D. Y.; and Lee, K. H. 2019. Arbitrary Style Transfer with Style-Attentional Networks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.
- Radford, A.; Kim, J. W.; Hallacy, C.; Ramesh, A.; Goh, G.; Agarwal, S.; Sastry, G.; Askell, A.; Mishkin, P.; Clark, J.; Krueger, G.; and Sutskever, I. 2021. Learning Transferable Visual Models From Natural Language Supervision. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, 8748–8763.
- Shang, C.; Wang, Z.; Wang, H.; and Meng, X. 2025. SCSA: A Plug-and-Play Semantic Continuous-Sparse Attention for Arbitrary Semantic Style Transfer. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 13051–13060.
- Wright, M.; and Ommer, B. 2022. ArtFID: Quantitative Evaluation of Neural Style Transfer. *arXiv preprint arXiv:2207.12280*.
- Yeh, M.-C.; Tang, S.; Bhattad, A.; Zou, C.; and Forsyth, D. 2020. Improving Style Transfer with Calibrated Metrics. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 3160–3168.
- Zhang, S.; Kang, H.; Liu, Y.; Mei, F.; and Li, H. 2025. HSI: A Holistic Style Injector for Arbitrary Style Transfer. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 23433–23442.
- Zhang, Z.; Zhang, Q.; Xing, W.; Li, G.; Zhao, L.; Sun, J.; Lan, Z.; Luan, J.; Huang, Y.; and Lin, H. 2024. ArtBank: Artistic Style Transfer with Pre-trained Diffusion Model and Implicit Style Prompt Bank. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, 7396–7404.
- Zheng, S.; Gao, P.; Zhou, P.; and Qin, J. 2024. Puff-Net: Efficient Style Transfer with Pure Content and Style Feature Fusion Network. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 8059–8068.
- Zhou, X.; Zeng, T.; Xu, H.; Liu, X.; Zhao, J.; and Yao, Y. 2024. A Comprehensive Evaluation of Arbitrary Image Style Transfer Methods. *IEEE Transactions on Visualization and Computer Graphics*, 30(1): 1–19.

Reproducibility Checklist

This paper

- Includes a conceptual outline and/or pseudocode description of AI methods introduced: **Yes**.
- Clearly delineates statements that are opinions, hypothesis, and speculation from objective facts and results: **Yes**.
- Provides well marked pedagogical references for less-familiar readers to gain background necessary to replicate the paper: **Yes**.

Does this paper make theoretical contributions? Yes (design-grounding). The paper provides bounded formal analysis with

proofs and empirical checks: Theorem 1 and Theorem 2 analyze the active field/solver regime through a path-action decomposition for kinetic regularization and an $O(1/K)$ Euler discretization bound, while Theorem 3 analyzes a historical OT-style assignment variant as family-level background. Full proofs are in the supplement. These results justify the three-part design without claiming equivalence to a full Schrödinger Bridge solver.

Does this paper rely on one or more datasets? Yes.

- A motivation is given for why the experiments are conducted on the selected datasets: **Yes**.
- All novel datasets introduced in this paper are included in a data appendix: **NA**.
- All novel datasets introduced in this paper will be made publicly available upon publication of the paper with a license that allows free usage for research purposes: **NA**.
- All datasets drawn from the existing literature are accompanied by appropriate citations: **Yes**.
- All datasets drawn from the existing literature are publicly available: **Yes**.
- All datasets that are not publicly available are described in detail, with explanation why publicly available alternatives are not scientifically satisfying: **NA**.

Does this paper include computational experiments? Yes.

- This paper states the number and range of values tried per (hyper-)parameter during development of the paper, along with the criterion used for selecting the final parameter setting: **Partial**.
- Any code required for pre-processing data is included in the appendix: **No**.
- All source code required for conducting and analyzing the experiments is included in a code appendix: **No**.
- All source code required for conducting and analyzing the experiments will be made publicly available upon publication of the paper with a license that allows free usage for research purposes: **Partial**.
- All source code implementing new methods have comments detailing the implementation, with references to the paper where each step comes from: **Partial**.
- If an algorithm depends on randomness, then the method used for setting seeds is described in a way sufficient to allow replication of results: **Partial**.
- This paper specifies the computing infrastructure used for running experiments, including hardware/software details: **Partial**.
- This paper formally describes evaluation metrics used and explains the motivation for choosing these metrics: **Yes**.
- This paper states the number of algorithm runs used to compute each reported result: **Yes**.
- Analysis of experiments goes beyond single-dimensional summaries of performance to include measures of variation, confidence, or other distributional information: **Yes**.
- The significance of any improvement or decrease in performance is judged using appropriate statistical tests: **Partial** (selected confidence intervals and bootstrap summaries are reported, but not every manuscript comparison is presented as a closed significance claim).
- This paper lists all final (hyper-)parameters used for each model/algorithm in the paper's experiments: **Partial**.